CS1231S Discrete Structures

WEEK 8 RECITATION
LECTURE 9. CARDINALITY

8 October 2026

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CS1231S (AY2026/27 Sem1) QnA





To infinity and beyond!

~ Buzz Lightyear



Which of the following statements are true?



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1. $|\{a, b, c\}| = |\{a, b\}|$
2. $|\mathbb{Z}^-| = |\mathbb{Z}^+|$
3. $|\mathbb{Z} \cup \{a, b, c\}| = |\mathbb{Z}|$
4. $|\mathbb{Z}| = |\mathbb{Z}^+|$
5. $|\mathbb{Z}| = |\{x \in \mathbb{Z} : x = 2k \text{ for some integer } k\}|$
6. $|\mathbb{Q}| = |\mathbb{Z}|$

Which of the following statements are true?



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✓2. $|\mathbb{Z}^-| = |\mathbb{Z}^+|$

✓3. $|\mathbb{Z} \cup \{a, b, c\}| = |\mathbb{Z}|$



✓4. $|\mathbb{Z}| = |\mathbb{Z}^+|$

✓5. $|\mathbb{Z}| = |\{x \in \mathbb{Z} : x = 2k \text{ for some integer } k\}|$

✓6. $|\mathbb{Q}| = |\mathbb{Z}|$





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Cardinality

Definitions: Finite set and Infinite set

Let $\mathbb{Z}_n = \{1,2,3, \dots, n\}$, the set of positive integers from 1 to n.

A set S is said to be **finite** iff S is empty, or there exists a bijection from S to $\mathbb{Z}_n$ for some $n \in \mathbb{Z}^+$.

A set S is said to be **infinite** if it is not finite.

Definition: Cardinality

The **cardinality** of a finite set S, denoted $|S|$, is

(i) 0 if $S = \emptyset$, or

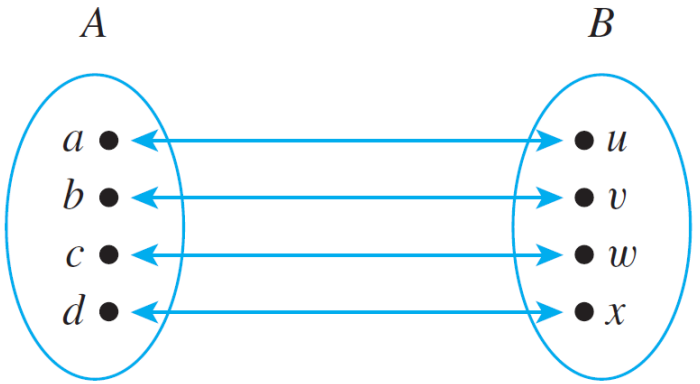
(ii) n if $f: S \to \mathbb{Z}_n$ is a bijection.

Theorem: Equality of Cardinality of Finite Sets

Let A and B be any finite sets.

$|A| = |B|$ iff there is a bijection $f: A \to B$.

We say that two finite sets whose elements can be paired by a bijection have the *same size*.



The elements of set A can be put into a bijection with the elements of B.

Cardinality

Theorem Cardinality.1: Subset of a Finite Set

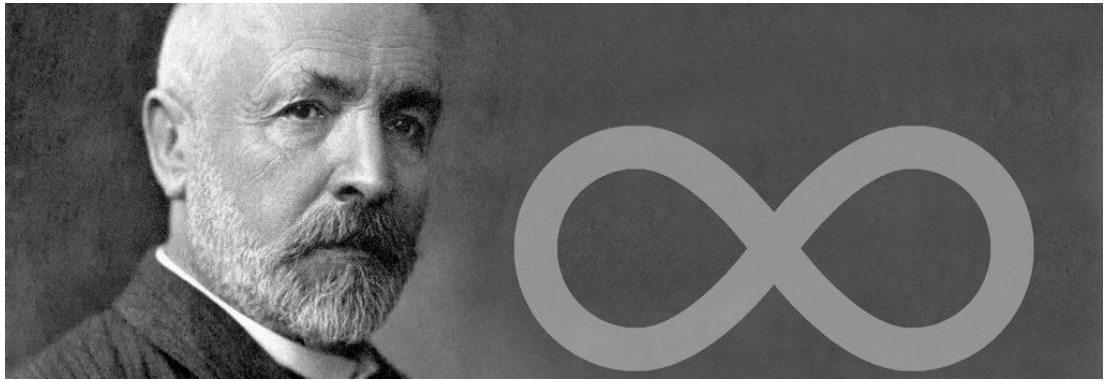
Any subset of a finite set is finite.
That is, let $A \subseteq B$. If B is finite, then A is finite.

The contrapositive of this theorem is:

Let $A \subseteq B$. If A is infinite, then B is infinite.

Cardinality

What about infinite sets?



Georg Cantor
(1845 – 1918)

https://en.wikipedia.org/wiki/Georg_Cantor

Definition: Same Cardinality (Cantor)

Given any two sets A and B. A is said to have the **same cardinality** as B, written as $|A| = |B|$, iff there is a bijection $f: A \to B$.



We define what $|A| = |B|$ means
without defining what $|A|$ and $|B|$ mean!

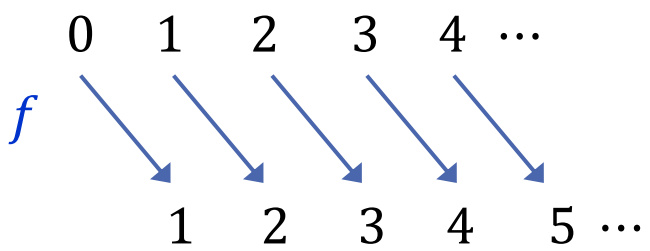
Cardinality

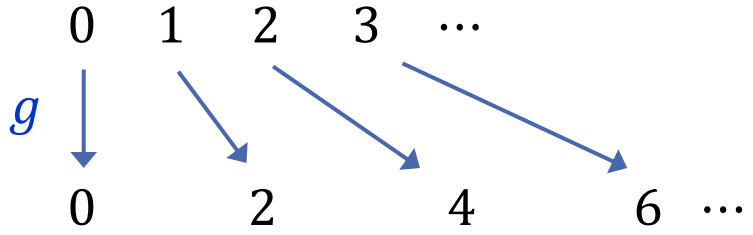
Definition: Same Cardinality (Cantor)

Given any two sets A and B. A is said to have the **same cardinality** as B, written as $|A| = |B|$, iff there is a bijection $f: A \to B$.

Example #1: $|\mathbb{N}| = |\ \mathbb{N} \setminus \{0\}|$ because the function $f: \mathbb{N} \to \mathbb{N} \setminus \{0\}$ satisfying $f(x) = x + 1$ for all $x \in \mathbb{N}$ is a bijection.

Recall we define $\mathbb{N} = \{0,1,2,3,\cdots\}$.





Example #2: $|\mathbb{N}| = |\mathbb{N} \setminus \{1,3,5,7,9,\cdots\}|$ because the function $g: \mathbb{N} \to \mathbb{N} \setminus \{1,3,5,7,9,\cdots\}$ satisfying $g(x) = 2x$ for all $x \in \mathbb{N}$ is a bijection.

Cardinality

Theorem 7.4.1 Properties of Cardinality

The same-cardinality relation is an equivalence relation.

For all sets A, B and C:

a. **Reflexive**: $|A| = |A|$.
b. **Symmetric**: $|A| = |B| \rightarrow |B| = |A|$.
c. **Transitive**: $(|A| = |B|) \wedge (|B| = |C|) \rightarrow |A| = |C|$.

Proof (reflexivity): To prove $|A| = |A|$.

It suffices to show that id_A is a bijection $A \rightarrow A$.

1. id_A is injective because if $x_1, x_2 \in A$ such that $id_A(x_1) = id_A(x_2)$, then $x_1 = x_2$.
2. id_A is surjective because given any $x \in A$, we have $id_A(x) = x$.
3. Therefore $|A| = |A|$.

Cardinality

Theorem 7.4.1 Properties of Cardinality

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c. **Transitive**: $(|A| = |B|) \wedge (|B| = |C|) \rightarrow |A| = |C|$.

Proof (symmetry): To prove $|A| = |B| \rightarrow |B| = |A|$.

1. Suppose $|A| = |B|$.
2. Use Cantor's definition of same-cardinality to find a bijection $f: A \rightarrow B$.
3. By Theorem 7.2.3, $f^{-1}: B \rightarrow A$ is also a bijection.
4. Therefore, $|B| = |A|$.

Theorem 7.2.3

If $f: X \rightarrow Y$ is a bijection, then $f^{-1}: Y \rightarrow X$ is also a bijection.
In other words, $f: X \rightarrow Y$ is bijective iff f has an inverse.

Cardinality

Theorem 7.4.1 Properties of Cardinality

The same-cardinality relation is an equivalence relation.

For all sets A, B and C:

a. **Reflexive**: $|A| = |A|$.
b. **Symmetric**: $|A| = |B| \rightarrow |B| = |A|$.
c. **Transitive**: $(|A| = |B|) \wedge (|B| = |C|) \rightarrow |A| = |C|$.

Proof (transitivity): To prove $(|A| = |B|) \wedge (|B| = |C|) \rightarrow |A| = |C|$.

1. Suppose $|A| = |B|$ and $|B| = |C|$.
2. Use Cantor's definition of same-cardinality to find a bijection $f: A \rightarrow B$ and a bijection $g: B \rightarrow C$.
3. From Tutorial 6 Q3, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
4. In particular, this means $g \circ f$ has an inverse.
5. So by Theorem 7.2.3, $g \circ f$ is a bijection $A \rightarrow C$.
6. Therefore, $|A| = |C|$.

Which of the following statements are true?

1. $|\{a, b, c\}| = |\{a, b\}|$
2. ✓ $|\mathbb{Z}^-| = |\mathbb{Z}^+|$
3. ✓ $|\mathbb{Z} \cup \{a, b, c\}| = |\mathbb{Z}|$
4. ✓ $|\mathbb{Z}| = |\mathbb{Z}^+|$
5. ✓ $|\mathbb{Z}| = |\{x \in \mathbb{Z} : x = 2k \text{ for some integer } k\}|$
6. ✓ $|\mathbb{Q}| = |\mathbb{Z}|$

Note that $\mathbb{Z}^+$ is a proper subset of $\mathbb{Z}$, that is, $\mathbb{Z}^+ \subseteq \mathbb{Z}$ and $\mathbb{Z}^+ \neq \mathbb{Z}$. And yet $|\mathbb{Z}^+| = |\mathbb{Z}|$! How strange!





For a finite set A, any proper subset B of A will have $|B| < |A|$. But this is not true for infinite sets.

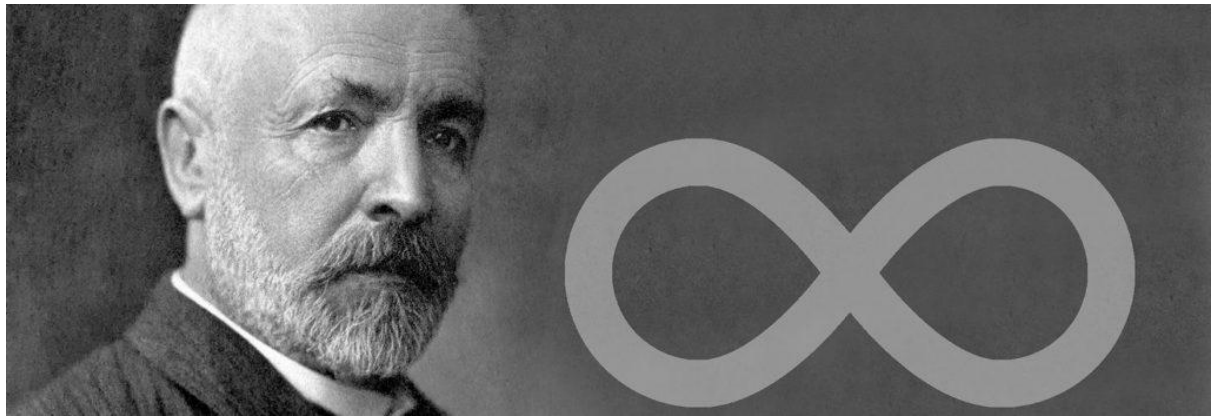
Some mathematicians have proposed to use this as the definition of an infinite set. That is, a set A is infinite iff there exists a set B such that $(B \subseteq A) \wedge (B \neq A) \wedge (|B| = |A|)$.

“No one shall drive us from the paradise which Cantor has created for us.”

~ David Hilbert.



David Hilbert
(1862 – 1943)



Georg Cantor (1845 – 1918)

Countably Infinite: Countable Sets

The set $\mathbb{Z}^+$ of counting numbers $\{1,2,3, \dots\}$ is in a sense, the most basic of all infinite sets.

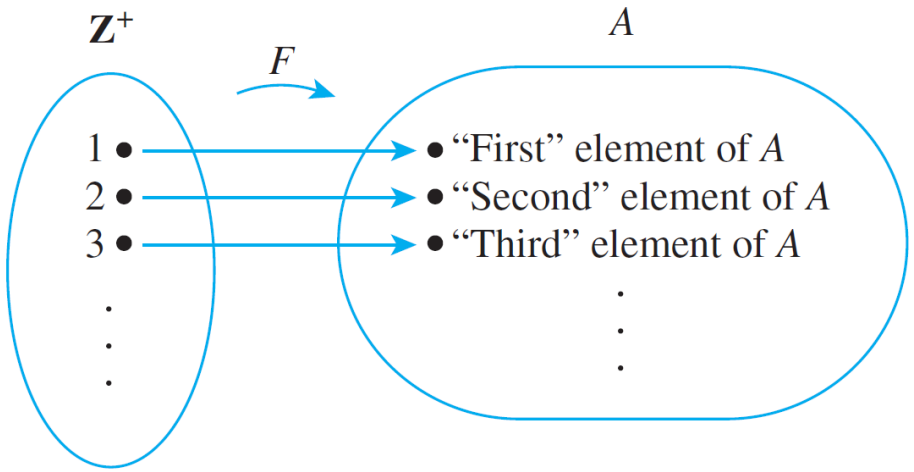
The set A having the same cardinality as $\mathbb{Z}^+$ is called countably infinite.

The reason is that the bijection between the two sets can be used to "count" the elements of A:

If F is the bijection from $\mathbb{Z}^+$ to A, then "$F(1)$" can be designated as the first element of A, "$F(2)$" as the second element of A, and so forth.

Note: We may use $\mathbb{N}$ or $\mathbb{Z}_{\geq 0}$ instead of $\mathbb{Z}^+$ in the definition.

Countably Infinite: Countable Sets



"Counting" a Countably Infinite Set
Figure 7.4.1

Because F is injective, every element of A is counted at most once; because F is surjective, every element of A is counted at least once.

Countably Infinite: Countable Sets

Definition: Cardinal numbers

Define $\aleph_0 = |\mathbb{Z}^+|$. (Some authors use $\mathbb{N}$ instead of $\mathbb{Z}^+$.)

$\aleph$ is pronounced "aleph", the first letter of the Hebrew alphabet. This is the first transfinite cardinal number.

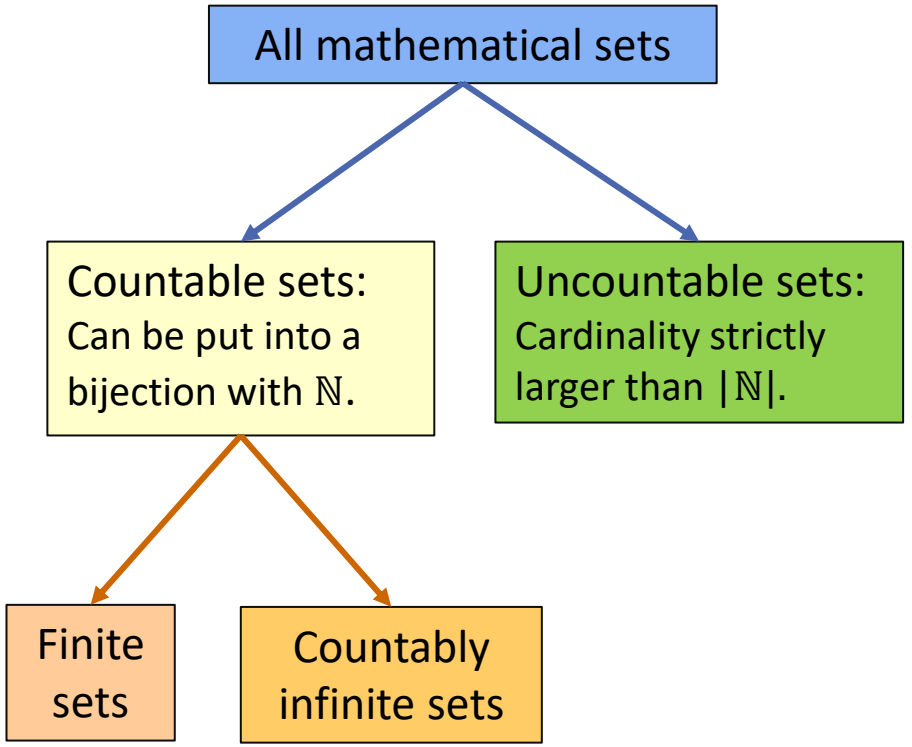
Definition: Countably infinite

A set S is said to be **countably infinite** (or, S has the cardinality of natural numbers) iff $|S| = \aleph_0$.

Definitions: Countable set and Uncountable set

A set is said to be **countable** iff it is finite or countably infinite.

A set is said to be **uncountable** if it is not countable.



Countably Infinite: $\mathbb{Z}$ is countable

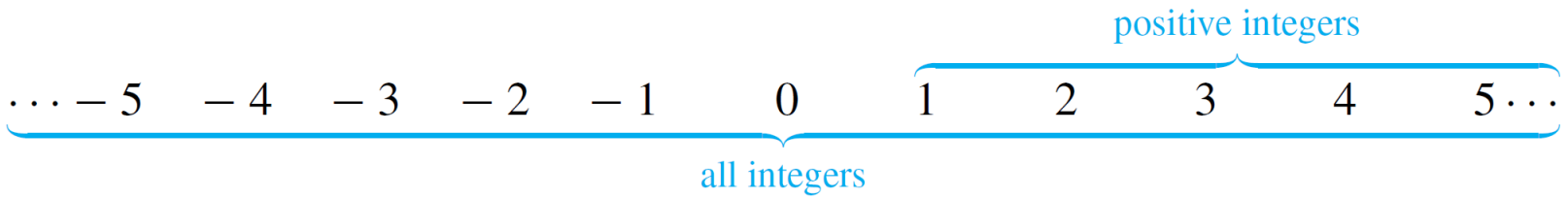
Example #3: Show that $\mathbb{Z}$ is countable.

(Intuitively, $\mathbb{Z}$ contains twice as many numbers as $\mathbb{Z}^+$, so we expect $|\mathbb{Z}| = 2|\mathbb{Z}^+|$. But this is not the case. They have the same cardinality!)

The set $\mathbb{Z}$ is certainly not finite, so if it is countable, it has to be countably infinite.

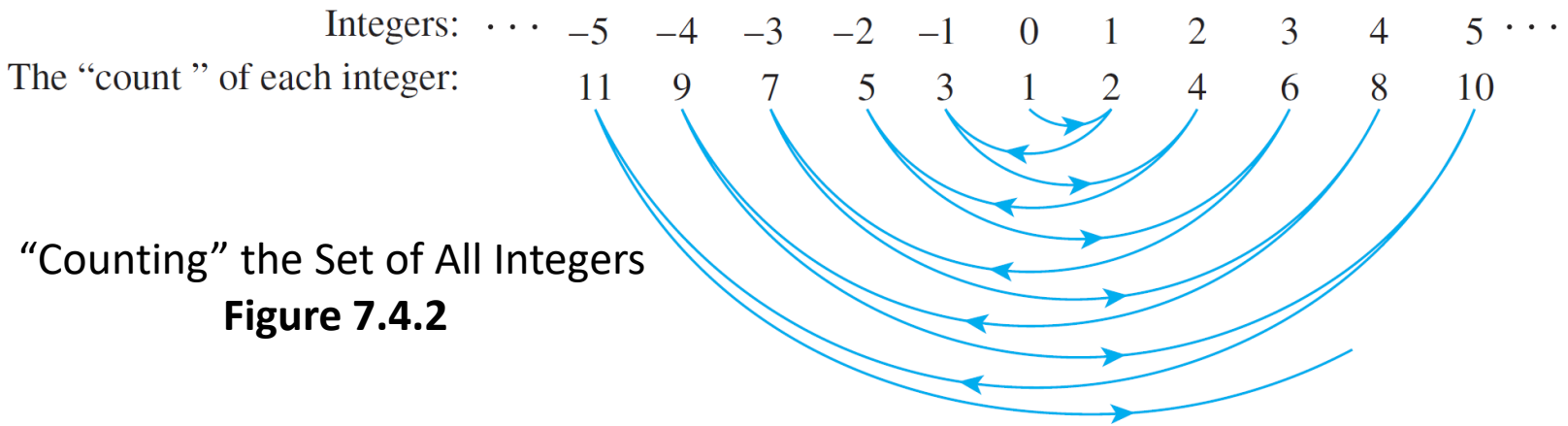
To show that $\mathbb{Z}$ is countably infinite, find a bijection from $\mathbb{Z}^+$ to $\mathbb{Z}$.

The appears to contradict common sense, as there appear to be more than twice as many integers as there are positive integers:



Countably Infinite: $\mathbb{Z}$ is countable

The trick is to start in the middle and work outward systematically. Let the first integer be 0, the second 1, the third -1, the fourth 2, the fifth -2, and so forth:



"Counting" the Set of All Integers
Figure 7.4.2

Every integer in $\mathbb{Z}$ is counted at most once (so the function is injective) and every integer in $\mathbb{Z}$ is counted at least once (so the function is surjective).

Therefore $\mathbb{Z}$ is countably infinite and hence countable.

Countably Infinite: $\mathbb{Z}$ is countable

The bijection from $\mathbb{Z}^+$ to $\mathbb{Z}$ can be formulated as follows:

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is an even positive integer} \\ -(n-1)/2, & \text{if } n \text{ is an odd positive integer} \end{cases}$$

$f(1) = -(1-1)/2 = 0$

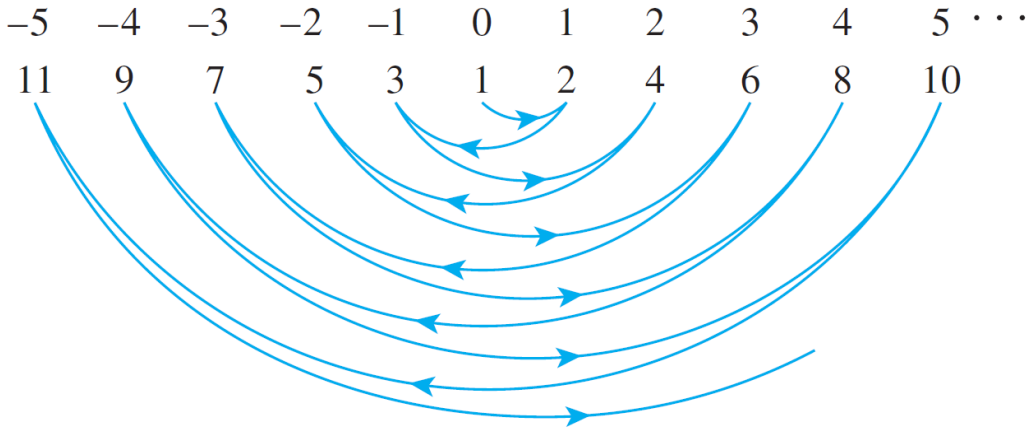
$f(2) = 2/2 = 1$

$f(3) = -(3-1)/2 = -1$

$f(4) = 4/2 = 2$

$f(5) = -(5-1)/2 = -2$

etc.



Countably Infinite: $\mathbb{Q}^+$ is countable

Example #4: Show that $\mathbb{Q}^+$ (the set of all positive rational numbers) is countable.

Display the elements of $\mathbb{Q}^+$ in a grid as shown:

Define a function F from $\mathbb{Z}^+$ to $\mathbb{Q}^+$ by starting to count at $\frac{1}{1}$ and following the arrows as indicated, skipping over any number that has already been counted.

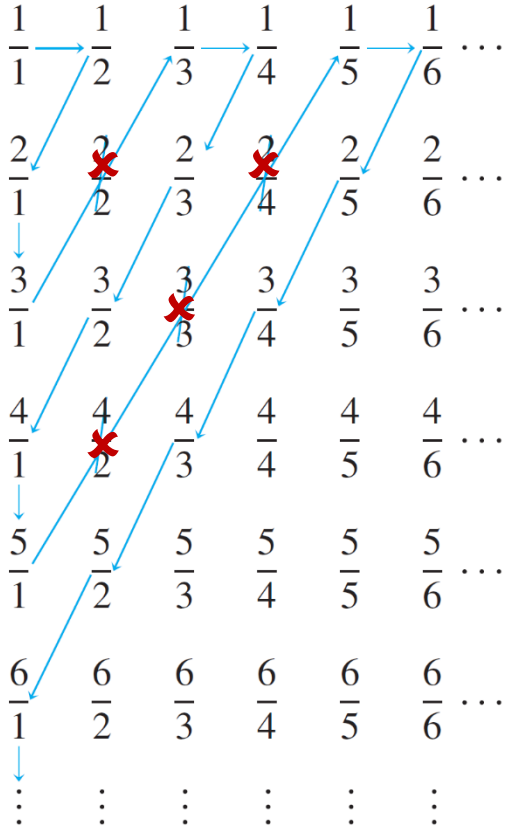


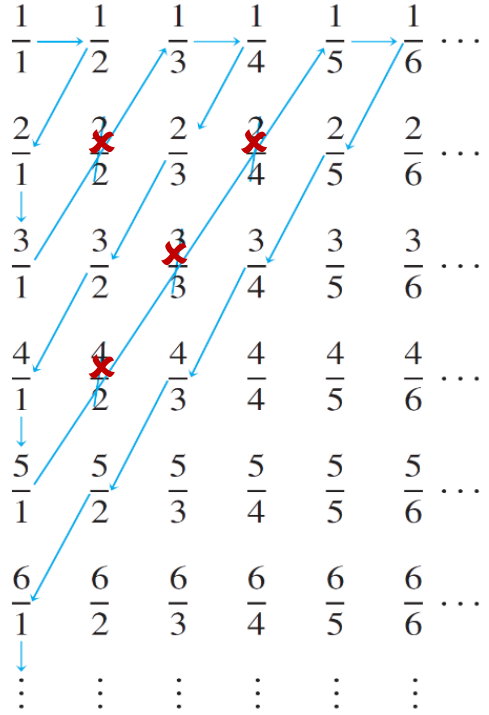
Figure 7.4.3

Countably Infinite: $\mathbb{Q}^+$ is countable

So, set $F(1) = \frac{1}{1}, F(2) = \frac{1}{2}, F(3) = \frac{2}{1}, F(4) = \frac{3}{1}$.

Then skip $\frac{2}{2}$ since $\frac{2}{2} = \frac{1}{1}$ which was counted.

Followed by $F(5) = \frac{1}{3}, F(6) = \frac{1}{4}, F(7) = \frac{2}{3}$, etc.



Note that every positive rational number appears somewhere in the grid, and the counting procedure is set up so that every point in the grid is reached eventually. Thus F is surjective.

Skipping numbers that have already been counted ensures that no number is counted twice. Thus F is injective.

So F is a bijection from $\mathbb{Z}^+$ to $\mathbb{Q}^+$. Therefore $\mathbb{Q}^+$ is countably infinite and hence countable.

Countably Infinite: $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable



The Infinite Hotel

Display the elements of $\mathbb{Z}^+ \times \mathbb{Z}^+$ in a grid as shown:

The ordered pair (x, y) denotes bus x and guest y.

We then count the ordered paired in the following order according to this function $f: \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Z}^+$ by:

Bus \ Guests	1	2	3	4	
1	(1,1)	(1,2)	(1,3)	(1,4)	⋯
2	(2,1)	(2,2)	(2,3)	(2,4)	⋯
3	(3,1)	(3,2)	(3,3)	(3,4)	⋯
4	(4,1)	(4,2)	(4,3)	(4,4)	⋯
⋮	⋮	⋮	⋮	⋮	

$$f(x, y) = \frac{(x + y - 2)(x + y - 1)}{2} + x$$

Countably Infinite: Theorems

Theorem (Cartesian Product)

If sets A and B are both countably infinite, then so is $A \times B$.

Corollary (General Cartesian Product)

Given $n \geq 2$ countably infinite sets $A_1, A_2, \cdots, A_n$, the Cartesian product $A_1 \times A_2 \times \cdots \times A_n$ is also countably infinite.

Theorem (Unions)

The union of countably many countable sets is countable. That is, if $A_1, A_2, \cdots$ are all countable sets, then so is

$$\bigcup_{i=1}^{\infty} A_i$$

Countability via Sequences

Definition: A set is said to be **countable** iff it is finite or countably infinite, that is, it has the same cardinality as $\mathbb{Z}_{\geq 0}$.

Note: In this section, we use $\mathbb{Z}_{\geq 0}$ instead of $\mathbb{Z}^+$ as we are relating to sequences.

Proposition 9.1

An infinite set B is countable if and only if there is a sequence $b_0, b_1, b_2, \cdots \in B$ in which every element of B appears exactly once.

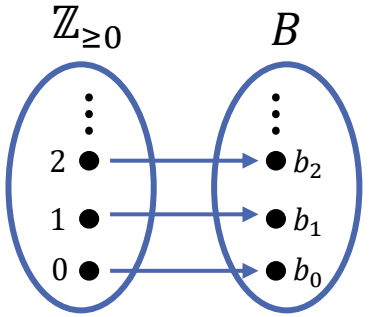
Justification:

function $f: \mathbb{Z}_{\geq 0} \to B$

$$f(0), f(1), f(2), \cdots$$

(surjectivity) $\forall b \in B\ \exists i \in \mathbb{Z}_{\geq 0}\ f(i) = b$

(injectivity) $\forall i, j \in \mathbb{Z}_{\geq 0}\ (f(i) = f(j) \Rightarrow i = j)$



sequence of elements of B

$$b_0, b_1, b_2, \cdots$$

Every $b \in B$ appears at least once.

Every $b \in B$ appears at most once.

Definition: A **sequence** $a_0, a_1, a_2, \cdots$ can be represented by a function a whose domain is $\mathbb{Z}_{\geq 0}$ that satisfies $a(n) = a_n$ for every $n \in \mathbb{Z}_{\geq 0}$.

Countability via Sequences

Proposition 9.1

An infinite set B is countable if and only if there is a sequence $b_0, b_1, b_2, \cdots \in B$ in which every element of B appears exactly once.

Lemma 9.2: Countability via Sequence

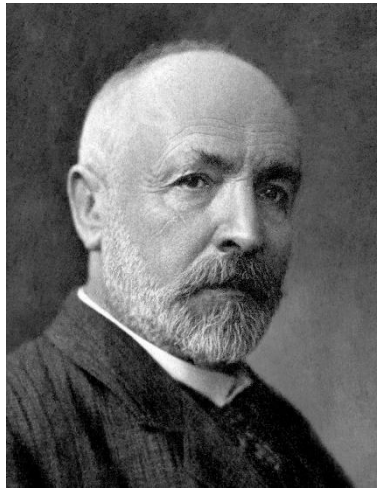
An infinite set B is countable if and only if there is a sequence $b_0, b_1, b_2, \cdots$ in which every element of B appears.

Proof:

1. ("only if") This follows directly from Proposition 9.1.
2. ("if")
   - 2.1. Let $b_0, b_1, b_2, \cdots$ be a sequence in which every element of B appears.
   - 2.2. Remove those terms in the sequence that are not in B.
   - 2.3. If an element of B appears more than once, then remove all but the first appearance.
   - 2.4. The result is a sequence in which every element of B appears exactly once.
   - 2.5. So B is countable.

Larger Infinities

In 1874 the German mathematician Georg Cantor achieved success in the search for a larger infinity by showing that the set of all real numbers is uncountable. His method of proof was somewhat complicated, however.



Georg Cantor, the man who discovered different infinities.

The uncountability of the set of all real numbers between 0 and 1 using a simpler technique introduced by Cantor in 1891 is known as the **Cantor's diagonalization process.**

Over the intervening years, this technique and variations on it have been used to establish a number of important results in logic and the theory of computation.

Cantor's Diagonalization Argument

Theorem 7.4.2 (Cantor)

The set of real numbers between 0 and 1,

$$(0,1) = \{x \in \mathbb{R} \mid 0 < x < 1\}$$

is uncountable.

To prove that a set is uncountable means proving that there is no possibility of a bijection from that set to $\mathbb{Z}^+$.

The way to prove is by contradiction. Georg Cantor, who gave us set theory, also gave an ingenious proof for the above theorem, came to be known as the Cantor's Diagonalization Argument.

$(0,1)$ is uncountable

Lemma 9.2: Countability via Sequence

An infinite set B is countable if and only if there is a sequence $b_0, b_1, b_2, \cdots$ in which every element of B appears.

Sketch of proof (proof by contradiction):

1. Suppose $(0,1)$ is countable.
2. Since it is not finite, it is countably infinite.
3. We list the elements x_i of $(0,1)$ in a sequence as follows:

$$x_1 = 0.\, a_{11}\, a_{12} a_{13} \cdots a_{1n} \cdots$$
$$x_2 = 0.\, a_{21}\, a_{22} a_{23} \cdots a_{2n} \cdots$$
$$x_3 = 0.\, a_{31}\, a_{32} a_{33} \cdots a_{3n} \cdots$$
$$\vdots$$
$$x_n = 0.\, a_{n1}\, a_{n2} a_{n3} \cdots a_{nn} \cdots$$
$$\vdots$$

where each $a_{ij} \in \{0,1,\cdots,9\}$ is a digit.

$(0,1)$ is uncountable

4. Now, construct a number $d = 0.\, d_1\, d_2 d_3 \cdots d_n \cdots$ s.t.

$$d_n = \begin{cases} 1, & \text{if } a_{nn} \neq 1; \\ 2, & \text{if } a_{nn} = 1. \end{cases}$$

5. Note that $\forall n \in \mathbb{Z}^+, d_n \neq a_{nn}$. Thus, $d \neq x_n, \forall n \in \mathbb{Z}^+$.
6. But clearly, $d \in (0,1)$, hence a contradiction. Therefore $(0,1)$ is uncountable.

$$\begin{aligned} x_1 &= 0.\, a_{11}\, a_{12} a_{13} \cdots a_{1n} \cdots \\ x_2 &= 0.\, a_{21}\, a_{22} a_{23} \cdots a_{2n} \cdots \\ x_3 &= 0.\, a_{31}\, a_{32} a_{33} \cdots a_{3n} \cdots \\ &\vdots \\ x_n &= 0.\, a_{n1}\, a_{n2} a_{n3} \cdots a_{nn} \cdots \end{aligned}$$

Illustration:

$0.\mathbf{2}0148802 \ldots$	d_1 is 1 because $a_{11} = 2$
$0.1\mathbf{1}666021 \ldots$	d_2 is 2 because $a_{22} = 1$
$0.03\mathbf{8}53320 \ldots$	d_3 is 1 because $a_{33} = 8$
$0.9677\mathbf{6}809 \ldots$	d_4 is 1 because $a_{44} = 7$
$0.00031\mathbf{1}002 \ldots$	d_5 is 2 because $a_{55} = 1$

Hence $d = 0.12112 \ldots$, which is not in the list. So, the list is incomplete. This is true regardless of how the elements in (0,1) are listed.

FAQ Hey, can't I use the same diagonalization argument to prove that positive integers are uncountable?

Suppose we list the positive integers in decimal notation:
1
2
3
:

To make them look like the real-number case, pad them with leading 0s:
…000001
…000002
…000003
:

Suppose we change one digit from each integer along a diagonal.
The resulting object will typically have infinitely many nonzero digits extending to the left, like …3141592653
But this is not an integer in ordinary decimal notation.

So the diagonalization argument may indeed construct something that isn't on the list, but it has done so by constructing something that is not even an integer!

In essence, a real number in (0,1) can have infinitely many digits: $0.d_1d_2d_3d_4\cdots$

An integer, however, has a finite decimal representation: $d_n d_{n-1}\cdots d_2\, d_1$.

The set of all finite binary strings $\{\varepsilon, 0,1,00,01,10,11,000,001,\dots\}$ is countable, while the set of all infinite binary strings $b_1b_2b_3b_4\cdots$, where $b_i \in \{0,1\}$ is uncountable.

Larger Infinities

Theorem 7.4.3

Any subset of any countable set is countable.

Proof:

1. Let A be any countable set and B be any subset of A.
2. If A is finite then B must be finite and hence countable – done.
3. Suppose A is countably infinite. If B is finite, then B is countable – done.
4. Suppose B is infinite.
   - 4.1 Since A is countable, there is a bijection $f: \mathbb{Z}^+ \to A$.
   - 4.2 Let $M = f^{-1}(B)$ (note that f^{-1} is a bijection), and define a function $g: \mathbb{Z}^+ \to B$ inductively as follows:
     - S1. Let $g(1) = f(i_1)$, where i_1 is the minimum element in M.
     - S2. If $g(1), g(2), \cdots, g(k-1)$ have been defined, …

Larger Infinities

Theorem 7.4.3

Any subset of any countable set is countable.

4. Suppose B is infinite.

   4.1 Since A is countable, there is a bijection $f: \mathbb{Z}^+ \to A$.

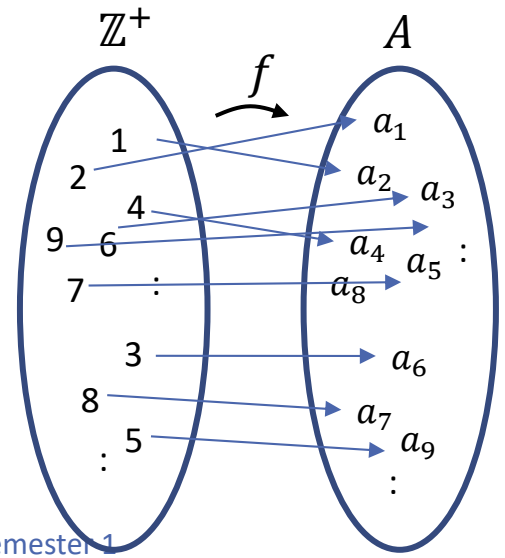
   4.2 Let $M = f^{-1}(B)$ (note that f^{-1} is a bijection), and define a function $g: \mathbb{Z}^+ \to B$ inductively as follows :

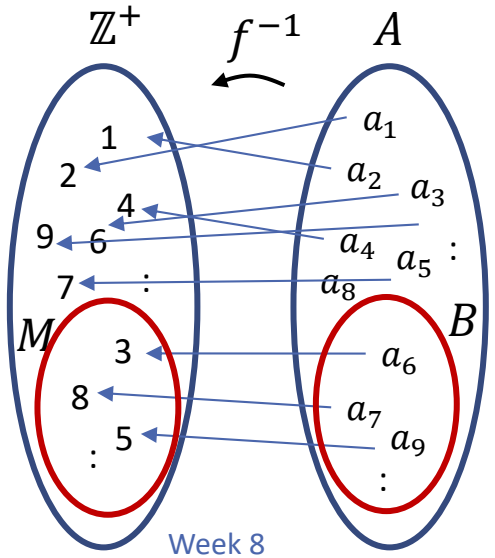
   S1. Let $g(1) = f(i_1)$, where i_1 is the minimum element in M.

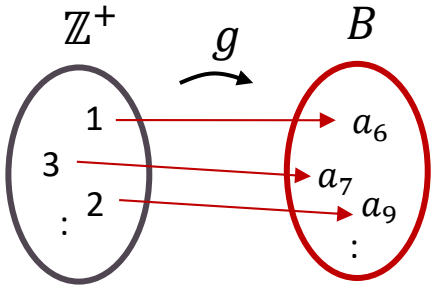
   S2. If $g(1), g(2), \cdots, g(k-1)$ have been defined, let

   $g(k) = f(i_k)$, where $i_k = \min\{m: m > i_{k-1}, m \in M\}$.

   4.3 g is a bijection (why?), hence B is countable.







Larger Infinities

Theorem 7.4.3

Any subset of any countable set is countable.

Corollary 7.4.4 (Contrapositive of Theorem 7.4.3)

Any set with an uncountable subset is uncountable.

Corollary 7.4.4 implies that $\mathbb{R}$ is uncountable since $(0,1) \subseteq \mathbb{R}$ and $(0,1)$ is uncountable.

Proposition 9.3

Every infinite set has a countably infinite subset.

Lemma 9.4: Union of Countably Infinite Sets.

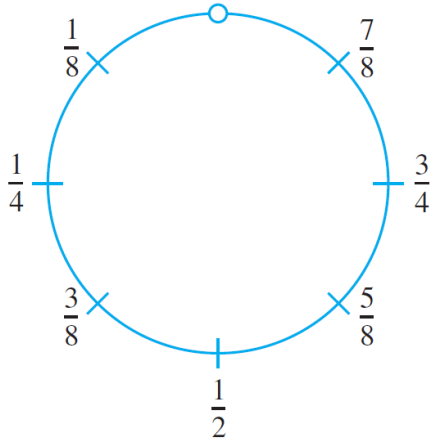
Let A and B be countably infinite sets. Then $A \cup B$ is countable.

Larger Infinities: Cardinality of $\mathbb{R}$

Example #5: Show that $|\mathbb{R}| = |(0,1)|$.

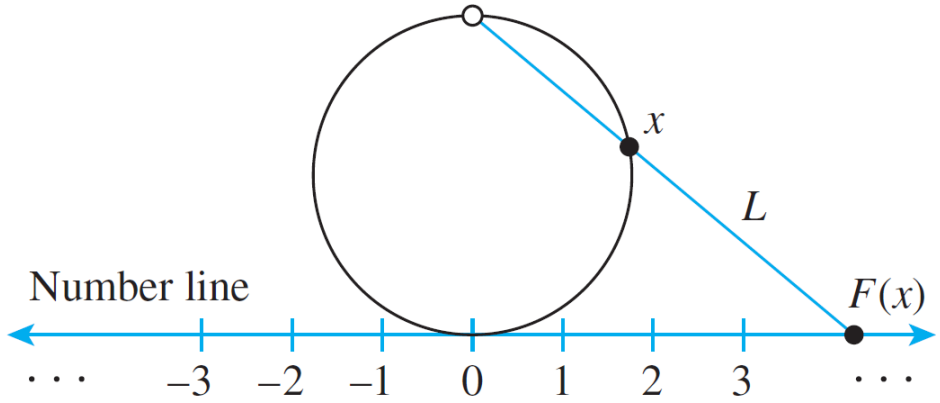
Let $S = (0,1)$, that is, $S = \{x \in \mathbb{R} \mid 0 < x < 1\}$.

Imagine picking up S and bending it into a circle:



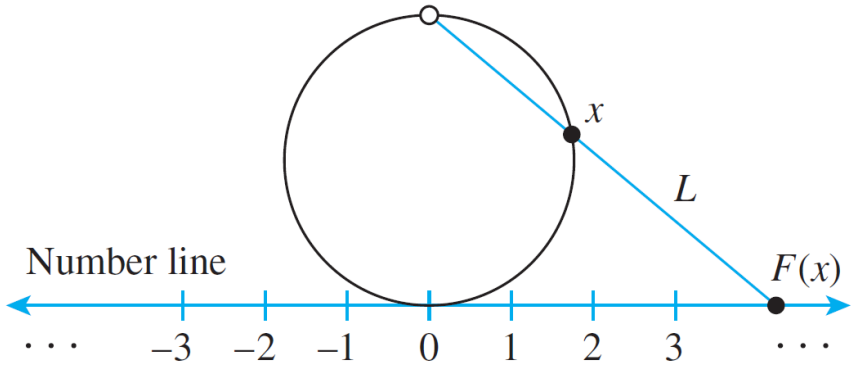
Define a function $F: S \to \mathbb{R}$ as follows:

Draw a number line and place the interval, S, bent into a circle as shown above, tangent to the line above the point 0, as shown below.



Larger Infinities: Cardinality of $\mathbb{R}$

For each point x on the circle representing S, draw a straight line L through the topmost point of the circle and x.



Let $F(x)$ be the point of intersection of L and the number line.

($F(x)$ is called the *projection of* x onto the number line.)

It can be seen that $F(x)$ is injective and surjective.

Hence S and $\mathbb{R}$ have the same cardinality, i.e. $|\mathbb{R}| = |(0,1)|$.

Why Study Cardinality in CS?

- Countability tells us what can, in principle, be represented by finite programs
  - Every computer program is a finite string over a finite alphabet.
  - Therefore, the set of all programs is countable.
- But there are uncountably many computational problems
  - Consider functions

    $$f : \mathbb{N} \to \{0,1\}.$$

    Every such function corresponds to an infinite binary sequence:

    $$f(0)f(1)f(2)f(3)\cdots$$

    The set of these functions is uncountable.

Cardinality gives us one of our first mathematical glimpses of the fundamental limits of computing.

Later CS courses:

- Theory of Computation
- Automata & Computability
- Algorithms/Complexity Theory
- Databases
- Cryptography/Security
- Probability/Randomized Algorithms
- etc.

Past-Year's Exam Question!

AY2022/23 Sem2

5. Two sets A and B are defined as follows:

$$A = \{n \in \mathbb{Z}^+ : (n > 1) \wedge \forall r, s \in \mathbb{Z}^+ \big(n = rs \rightarrow (r = n) \vee (r = 1)\big)\};$$
$$B = \{n \in \mathbb{Z}^+ : \exists r, s \in \mathbb{Z}^+ \big(n = rs \wedge (1 < r < n) \wedge (1 < s < n)\big)\}.$$

Which of the following are true?

(i) A and B are countable.

(ii) A is countable but B is uncountable.

(iii) $\mathcal{P}(A)$ is countable.

A. Only (i).

B. Only (ii).

C. Only (i) and (iii).

D. Only (ii) and (iii).

E. None of the above.

Both A and B are subset of $\mathbb{Z}^+$, hence they are countable (by Theorem 7.4.3).

$\mathcal{P}(A)$ is uncountable by Tutorial 8 Q7.

Past-Year's Exam Question!

AY2022/23 Sem1

17. Which of the following are countable sets?

(i) The set of relations on $\mathbb{Z}$.

(ii) The set of rational numbers between 0 and 1 exclusive.

(iii) The set $\mathbb{Z}^*$ of all finite strings over $\mathbb{Z}$.

A. Only (i).

B. Only (ii).

C. Only (i) and (ii).

D. Only (ii) and (iii).

E. All of (i), (ii) and (iii).

(i) A relation on $\mathbb{Z}$ is a subset of $\mathbb{Z}\times\mathbb{Z}$. The set of all relations on $\mathbb{Z}$ is therefore $\mathcal{P}(\mathbb{Z}\times\mathbb{Z})$.

(ii) $\mathbb{Q}$ is countable.

(iii) For each $n \geq 0$, let $\mathbb{Z}^n$ be the set of strings of length n.
Thus $\mathbb{Z}^* = \mathbb{Z}^0 \cup \mathbb{Z}^1 \cup \mathbb{Z}^2 \cup \cdots$
We know that $\mathbb{Z}$ is countable, and so $\mathbb{Z}^n$ for $n \geq 0$ are countable (general cartesian product), and so

$$\bigcup_{n=0}^{\infty} \mathbb{Z}^n$$

is countable (Tutorial 8 Q5).

Theorem 7.4.3
Any subset of any countable set is countable.

Corollary (General Cartesian Product)
Given $n \geq 2$ countably infinite sets $A_1, A_2, \cdots, A_n$, the Cartesian product $A_1 \times A_2 \times \cdots \times A_n$ is also countably infinite.

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